

Semester Test 3

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Electricity and Electronics EBN 111

6 June 2023

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)			18

Lecturers: Prof X. Ye, Prof J.D. le Roux, Mr J. Thiselton**Assistant Lecturers:** G. Neeuwfan, S. Twala, B. Mahlakoana, M. Modiba**ECSA outcome assessment**

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2.

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.
8. A complex number may be written in either rectangular or polar format.

INSTRUCTIONS

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers to the OCR form.

- The Assessment ID is: **2023EBN111S03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Make use of full stops as decimal points.

Examples for complex numbers on OCR forms

5	∠	—	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
—	5	0	.	9	4	°				
1	2	—	j	8						
j	1	5							Ω	
—	1	.	5	+	j	8	.	2	m	V

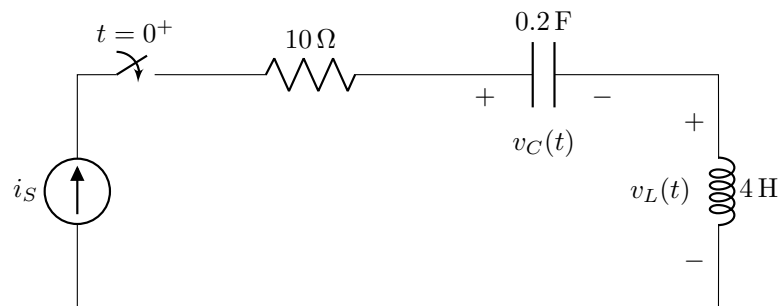
Rectangular:The j is at the start of the imaginary number.The j is in the CENTRE of the cell.The j does NOT cross or touch cell boundaries!**Polar:**

The degree sign ° is directly after the angle value.

Section 1

Questions 1 to 2 [4]

The capacitor and inductor are uncharged before the switch is closed at $t = 0^+$ s.
 If $i_S = (1 - e^{-t})$ A, determine the voltage across the capacitor $v_C(t)$ and inductor $v_L(t)$ respectively at $t = 1$ s.



01	$v_C(t = 1 \text{ s})$ [2]
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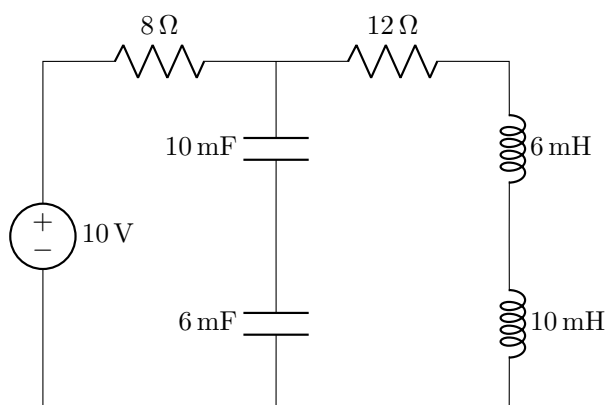
02	$v_L(t = 1 \text{ s})$ [2]
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$$\begin{aligned}
 * \quad v_C(1) &= \frac{1}{0.2} \int_0^1 (1 - e^{-t}) dt \\
 &= 5 [t + e^{-t}]_0^1 \\
 &= 5 [(1 + e^{-1}) - (0 + e^0)] \\
 &= 5 (1 + e^{-1} - 1) \\
 &= 5e^{-1} = \underline{1.839 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 * \quad v_L(1) &= 4 \frac{d}{dt} (1 - e^{-t}) \Big|_{t=1} \\
 &= 4 (e^{-t}) \Big|_{t=1} \\
 &= 4e^{-1} = \underline{1.472 \text{ V}}
 \end{aligned}$$

Questions 3 to 4 [4]

If the circuit is at steady-state, i.e., direct current conditions, determine the total energy stored by the capacitors $w_{C_{total}}$ and inductors $w_{L_{total}}$, respectively.



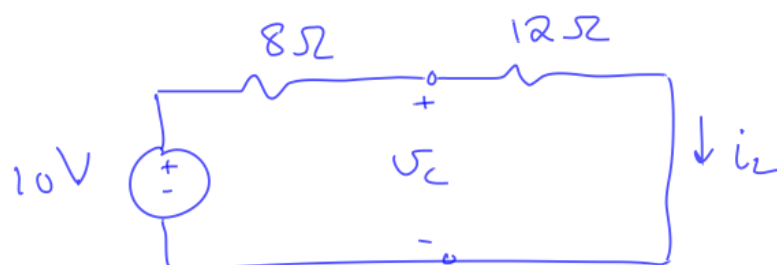
03 $w_{C_{total}}$ [2]

04 $w_{L_{total}}$ [2]

$$\rightarrow C_T = (10 \parallel 6) \text{ mF} = 3.75 \text{ mF}$$

$$L_T = (6 + 10) \text{ mH} = 16 \text{ mH}$$

$\rightarrow$ DC circuit:



$$* i_L = \frac{10}{20} = 0.5 \text{ A}$$

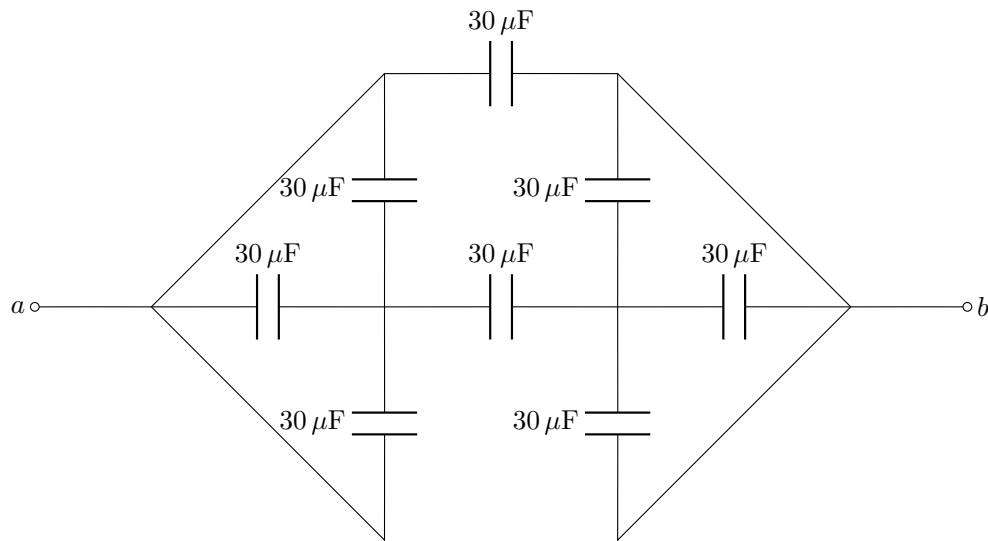
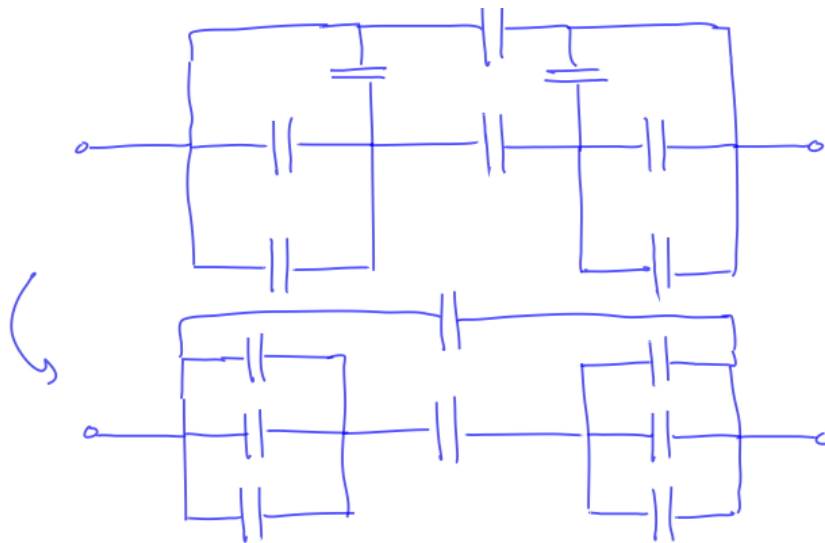
$$* V_C = \frac{10}{20} \times 12 = 6 \text{ V}$$

$$\therefore w_{C_T} = \frac{1}{2} (3.75 \text{ m}) (6)^2$$

$$= \underline{67.5 \text{ mJ}}$$

$$w_L = \frac{1}{2} (16 \text{ m}) (0.5)^2$$

$$= \underline{2 \text{ mJ}}$$

Question 5 [2]Determine the equivalent capacitance C_{ab} with respect to terminals a - b .05 | C_{ab} [2]

$$\begin{aligned}
 \therefore C_{ab} &= [3(30) \parallel 30 \parallel 3(30)] + 30 \\
 &= 90 \parallel 30 \parallel 90 + 30 \\
 &= \underline{48 \mu F}
 \end{aligned}$$

Total Section 1: [10]

Section 2

Question 6 [2]

Simplify the complex number $\mathbf{Z}$.

$$\mathbf{Z} = \frac{3 + j4 + 8\angle(-90^\circ)}{(2\angle 60^\circ)^3 + j4}$$

06	$\mathbf{Z} \text{ [2] } = -0.5 + j0.25$
----	--

$$\begin{aligned}
 \overline{\mathbf{Z}} &= \frac{3 + j4 + 8\angle -90^\circ}{(2^3\angle 60^\circ \times 3) + j4} \\
 &= \frac{3 + j4 + 8\angle -90^\circ}{(8\angle 180^\circ) + j4} \\
 &= \underline{\underline{-0.5 + j0.25}}
 \end{aligned}$$

Question 7 [2]

Determine $\mathbf{I}_1$.

$$\begin{pmatrix} 2\angle 30^\circ & -j4 \\ -j2 & 2+j4 \end{pmatrix} \begin{pmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} 8\angle 90^\circ \\ 0 \end{pmatrix}$$

07	$\mathbf{I}_1$ [2] $-0.7089 + j2.9915$
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$$\bar{\mathbf{I}}_1 = \frac{\Delta_1}{\Delta}, \text{ where } \Delta_1 = \begin{vmatrix} 8\angle 90^\circ & -j4 \\ 0 & 2+j4 \end{vmatrix}$$

$$\Rightarrow \Delta_1 = (8\angle 90^\circ)(2+j4) - (-j4)(0) \\ = -32 + j16$$

$$\Delta = \begin{vmatrix} 2\angle 30^\circ & -j4 \\ -j2 & 2+j4 \end{vmatrix}$$

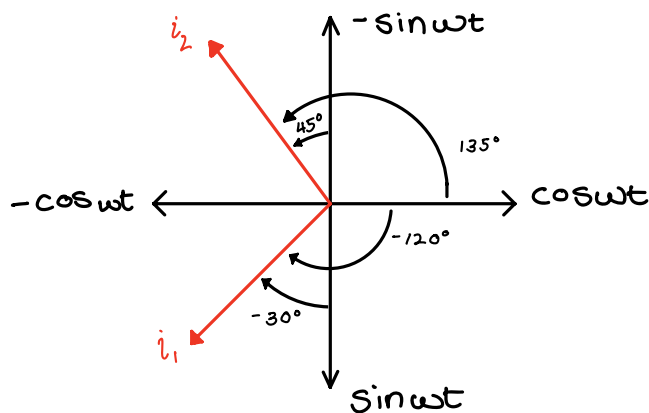
$$\Rightarrow \Delta = (2\angle 30^\circ)(2+j4) - (-j4)(-j2) \\ = 7.4641 + j8.928203$$

$$\Rightarrow \bar{\mathbf{I}}_1 = \frac{-32 + j16}{(7.4641 + j8.928203)} \\ = \underline{\underline{-0.7089 + j2.991517}}$$

Questions 8 to 9 [4]	
Given two sinusoidal signals: $i_1(t) = 10 \cos(\omega t - 120^\circ),$ $i_2(t) = -30 \sin(\omega t + 45^\circ).$	
08	Find the phase angle θ between $i_1(t)$ and $i_2(t)$. Write your answer in degrees. [2] $\theta = 105^\circ$
09	The signals $i_1(t)$ and $i_2(t)$ are represented by phasors $\mathbf{I}_1$ and $\mathbf{I}_2$, respectively. Calculate the phasor $\mathbf{I}_T = \mathbf{I}_1 - \mathbf{I}_2$. [2] $6.213 - j29.873$

$$i_1(t) = 10 \cos(\omega t - 120^\circ) \text{ A}$$

$$i_2(t) = -30 \sin(\omega t + 45^\circ) \text{ A} = 30 \cos(\omega t + 135^\circ) \text{ A}$$



$$\therefore \bar{\mathbf{I}}_1 = 10 \angle -120^\circ \text{ A}$$

$$\bar{\mathbf{I}}_2 = 30 \angle 135^\circ \text{ A}$$

$$\theta = 240^\circ - 135^\circ$$

$$\underline{\underline{\theta = 105^\circ}}$$

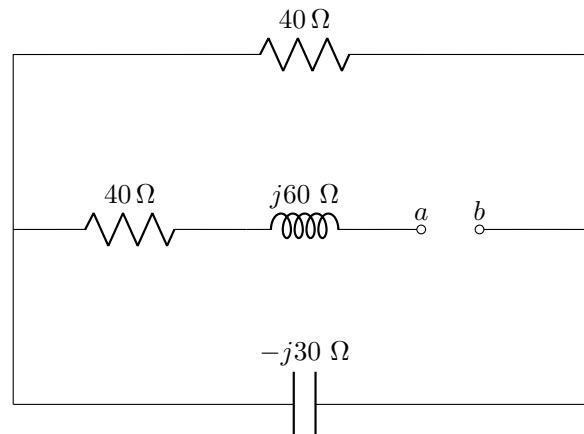
$$\bar{\mathbf{I}}_T = \bar{\mathbf{I}}_1 - \bar{\mathbf{I}}_2$$

$$\bar{\mathbf{I}}_T = 10 \angle -120^\circ - 30 \angle 135^\circ$$

$$\underline{\underline{\bar{\mathbf{I}}_T = 6.213 - j29.873 \text{ A}}}$$

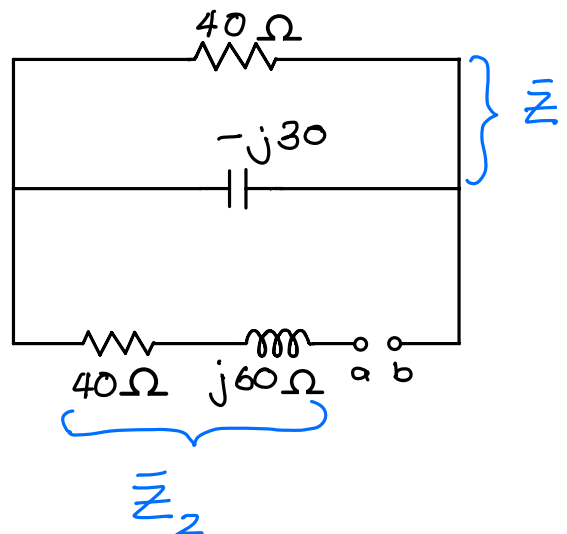
Question 10 [2]

Determine the total impedance of Z_{ab} with respect to terminals a - b .



10 Z_{ab} [2] $54.4 + j40.8 \Omega$

1) Rearrange circuit:



$$2) \bar{Z}_1 = (40 // -j30)$$

$$\frac{1}{\bar{Z}_1} = \frac{1}{40} + \frac{1}{-j30}$$

$$\Rightarrow \bar{Z}_1 = 14.4 - j19.2 \Omega$$

$$3) \bar{Z}_{ab} = \bar{Z}_1 + \bar{Z}_2$$

$$= (14.4 - j19.2) + (40 + j60)$$

$$\Rightarrow \bar{Z}_{ab} = \underline{\underline{54.4 + j40.8 \Omega}}$$

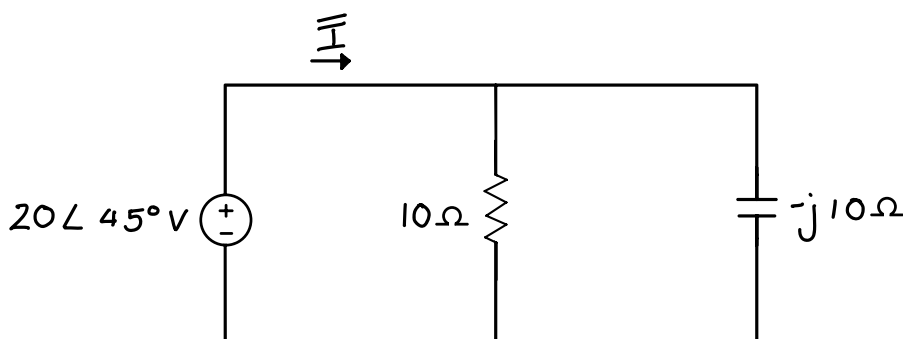
Question 11 [2]

If a voltage source $v(t) = 20\cos(500t + 45^\circ)$ V is connected in parallel with a $10\ \Omega$ resistor, and also in parallel with a $0.2\ \text{mF}$ capacitor. What is the phasor current $\mathbf{I}$ that exits the positive terminal of the voltage source?

11 | $\mathbf{I}$ [2] $j\ 2.828\ \text{A}$

$$v(t) = 20\cos(500t + 45^\circ) \Rightarrow 20\angle 45^\circ\ \text{V} = \bar{V}$$

① $C = 0.2\ \text{mF} \rightarrow \text{convert to impedance} \Rightarrow \bar{Z}_C = \frac{1}{j\omega C} = \frac{1}{j(500)(0.2 \times 10^{-3})} = j10\ \Omega$



$$\bar{Z} = \frac{10(-j10)}{10 - j10}$$

$$\bar{Z} = 5\sqrt{2}\angle 45^\circ\ \Omega$$

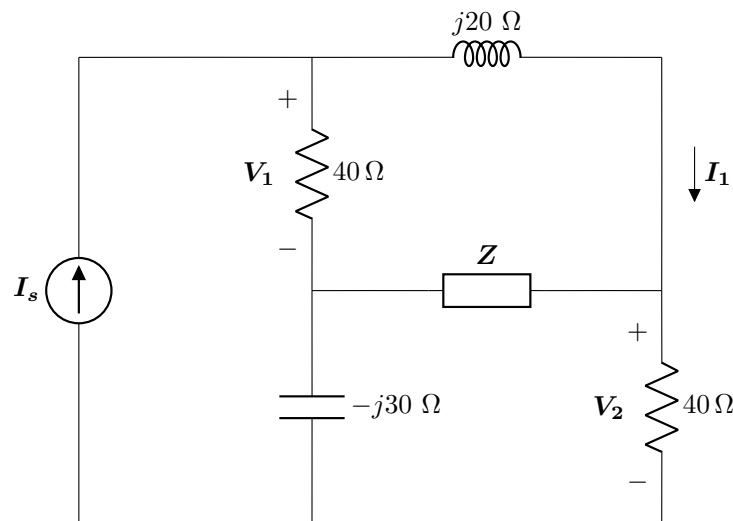
$$= 5 - j5\ \Omega$$

$$\therefore \bar{I} = \frac{\bar{V}}{\bar{Z}}$$

$$\bar{I} = \frac{20\angle 45^\circ}{5\sqrt{2}\angle 45^\circ} = \frac{20\angle 45^\circ}{5 - j5}$$

$$\bar{I} = \underline{\underline{j\ 2.8284\ \text{A}}}$$

Questions 12 to 13 [4]



12 If $Z = \infty\ \Omega$ and $I_s = 10\angle 0^\circ\text{ A}$, find I_1 . [2]

13 If $Z = 0\ \Omega$ and $V_1 = 80\angle 0^\circ\text{ V}$, find V_2 . [2]

$$12. I_1 = I_s \cdot \frac{40 - j30}{40 - j30 + 40 + j20} = 10 \cdot \frac{40 - j30}{80 - j10}$$

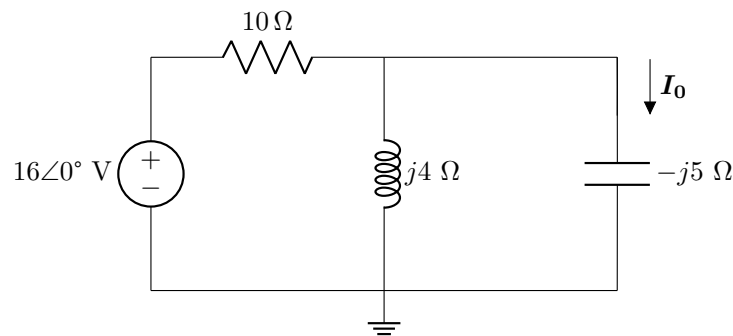
$$= 5.3846 - j3.0769$$

$$13. I = \frac{V_1}{40} + \frac{V_1}{j20} = 2 - j4\text{ (A)}$$

$$V_2 = I \cdot (40 \parallel -j30) = (2 - j4) \cdot \frac{40 \cdot (-j30)}{40 - j30}$$

$$= -48 - j96\text{ V.}$$

Question 14 [2]



14

Using an appropriate circuit analysis approach, determine I_0 [2] $-1.28 + j2.56 \text{ A}$

$$\textcircled{1} \quad \bar{Z}_1 = j4 \parallel -j5$$

$$\bar{Z}_1 = \frac{(j4)(-j5)}{j4 - j5}$$

$$\underline{\underline{\bar{Z}_1 = j20 \Omega}}$$

$$\textcircled{3} \quad \bar{V}_0 = \bar{V}_1 = 12.8 + j6.4 \text{ V}$$

$$\textcircled{2} \quad \bar{V}_1 = \frac{\bar{Z}_1}{\bar{Z}_1 + 10} \times 16 \angle 0^\circ$$

$$\bar{V}_1 = \frac{j20}{10 + j20} \times 16 \angle 0^\circ$$

$$\underline{\underline{\bar{V}_1 = 12.8 + j6.4 \text{ V}}}$$

$$\textcircled{4} \quad \bar{I}_0 = \frac{\bar{V}_0}{-j5}$$

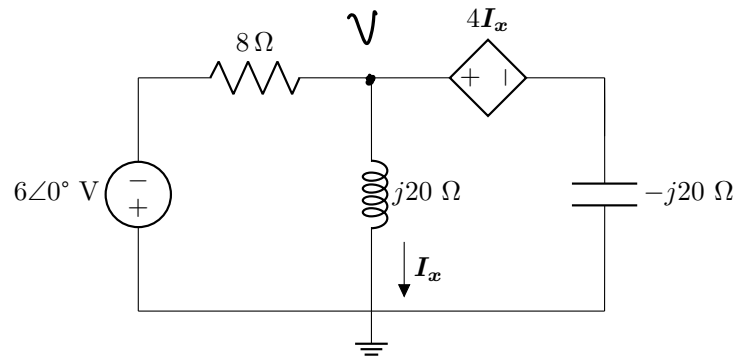
$$\bar{I}_0 = \frac{12.8 + j6.4}{-j5}$$

$$\underline{\underline{\bar{I}_0 = -1.28 + j2.56 \text{ A}}}$$

Total Section 2: [18]

Section 3

Question 15 [2]

15 Find I_x [2]

$$\frac{V+6}{8} + \frac{V}{j20} + \frac{V-4I_x}{-j20} = 0$$

$$\frac{V+6}{8} + \cancel{\frac{V}{j20}} + \cancel{\frac{V}{-j20}} + \frac{I_x}{j5} = 0$$

$$\frac{V+6}{8} + \frac{V/j20}{j5} = 0$$

$$\frac{V+6}{8} - \frac{V}{100} = 0$$

$$100(V+6) = 8V$$

$$\Rightarrow V = \frac{-600}{92} = -\frac{150}{23}$$

$$I_x = \frac{V}{j20} = -\frac{150}{j20 \times 23} = j0.3261$$

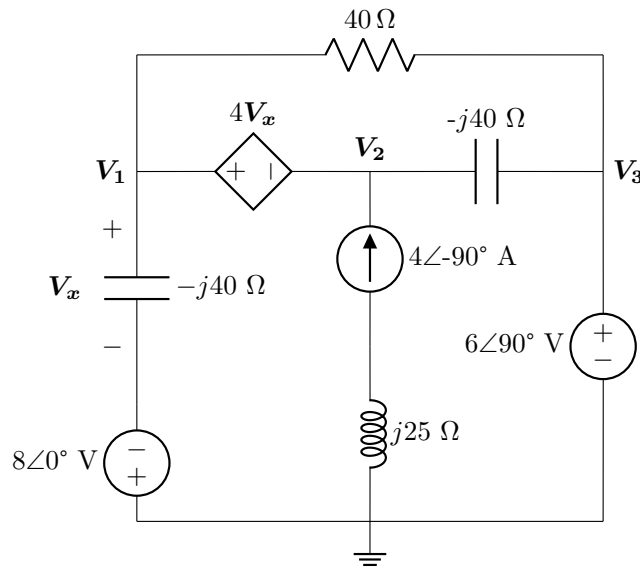
Questions 16 to 19 [4]

Use nodal analysis to setup two equations of the following form:

$$aV_1 + bV_2 = 32\angle 0^\circ,$$

$$cV_1 + jV_2 = d.$$

Find the complex coefficient of **a**, **b**, **c**, and **d**.



16	a [1]	-3
17	b [1]	-1
18	c [1]	1 + j
19	d [1]	-6 - j162

$$\bar{V}_x = \bar{V}_1 - (-8) = \bar{V}_1 + 8\angle 0^\circ$$

$$\bar{V}_1 - \bar{V}_2 = 4\bar{V}_x$$

$$\bar{V}_1 - \bar{V}_2 = 4[\bar{V}_1 + 8\angle 0^\circ]$$

$$\bar{V}_1 - \bar{V}_2 = 4\bar{V}_1 + 32\angle 0^\circ$$

$$\bar{V}_1(1-4) - \bar{V}_2 = 32\angle 0^\circ$$

$$-3\bar{V}_1 - \bar{V}_2 = 32\angle 0^\circ$$

$\bar{a}$	$= -3$
$\bar{b}$	$= -1$

$$\frac{\bar{V}_1 - 8}{-j40} + \frac{\bar{V}_1 - 6\angle 90^\circ}{40} + \frac{\bar{V}_2 - 6\angle 90^\circ}{-j40} - 4\angle -90^\circ = 0$$

$$j(\bar{V}_1 + 8) + \bar{V}_1 - j6 + j(\bar{V}_2 - j6) + j160 = 0$$

$$\bar{V}_1(j+1) + j\bar{V}_2 + j8 - j6 + 6 + j160 = 0$$

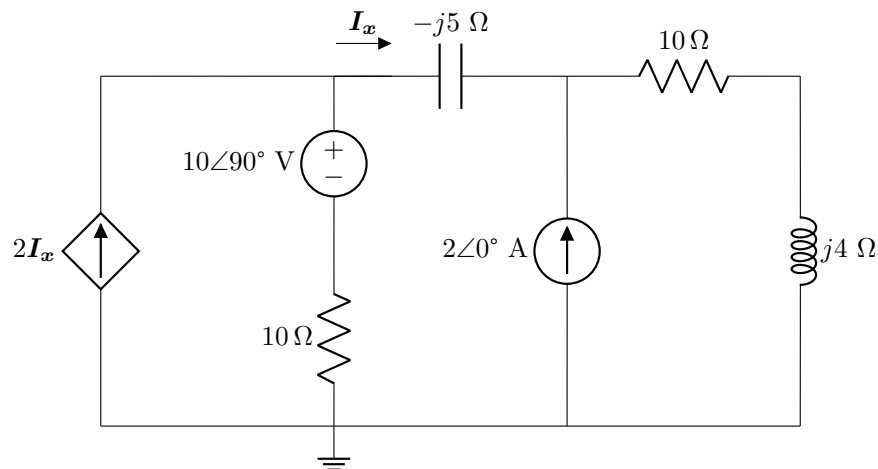
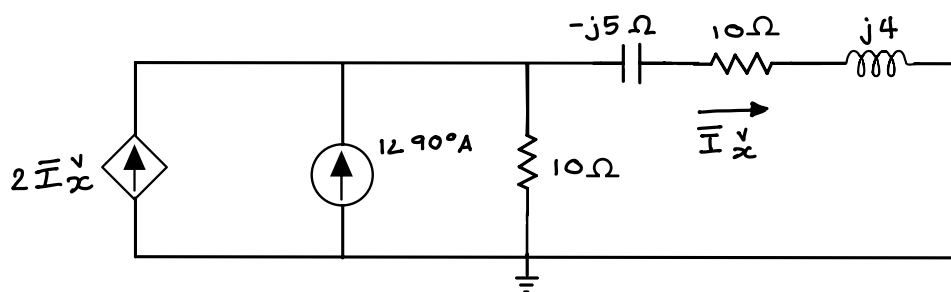
$$(j+1)\bar{V}_1 + j\bar{V}_2 + j162 + 6 = 0$$

$$\underline{(j+1)\bar{V}_1 + j\bar{V}_2 = -6 - j162}$$

$$\therefore c = j+1 = 1+j$$

$$d = -6 - j162$$

Question 20 [2]

Determine the exclusive contribution I_x^V of the independent voltage source to I_x .20 I_x^V [2] -10 A 

$$\bar{I}_x = \frac{\bar{V}_x - 0}{10 - j5 + j4}$$

$$\bar{I}_x = \frac{\bar{V}_x}{10 - j1}$$

$$2\bar{I}_x + j - \frac{\bar{V}_x - 0}{10} - \bar{I}_x = 0 \quad (2)$$

$$\bar{V}_x = (10 - j)\bar{I}_x \quad (1)$$

$$(1) \text{ into } (2) : 2\bar{I}_x + j - \frac{(10 - j)\bar{I}_x}{10} - \bar{I}_x = 0$$

$$20\bar{I}_x + j10 - (10 - j)\bar{I}_x$$

$$20\bar{I}_x - 10\bar{I}_x + j\bar{I}_x - 10\bar{I}_x + j10 = 0$$

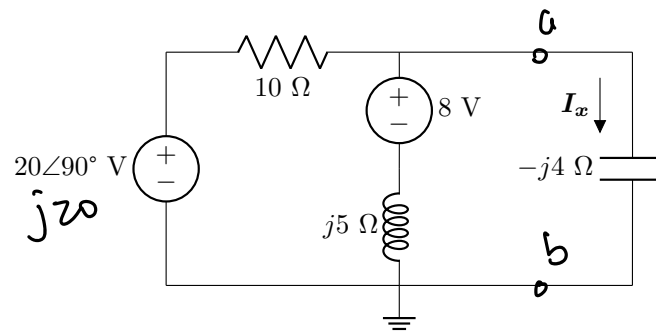
$$j\bar{I}_x = -j10$$

$$\underline{\underline{\bar{I}_x = -10\text{ A}}}$$

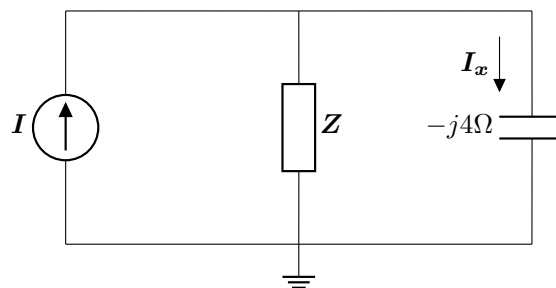
Questions 21 to 22 [4]

Use source transformation to transform Circuit A to Circuit B. Determine the current I and the impedance Z in Circuit B.

Circuit A:

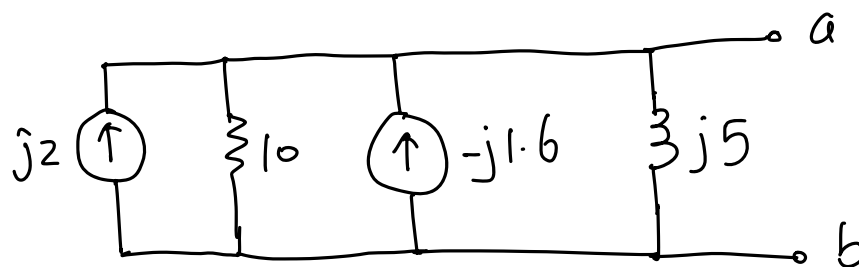


Circuit B:



21 I [2] $j0.4 \text{ A}$

22 Z [2] $2 + j4 \text{ } \Omega$

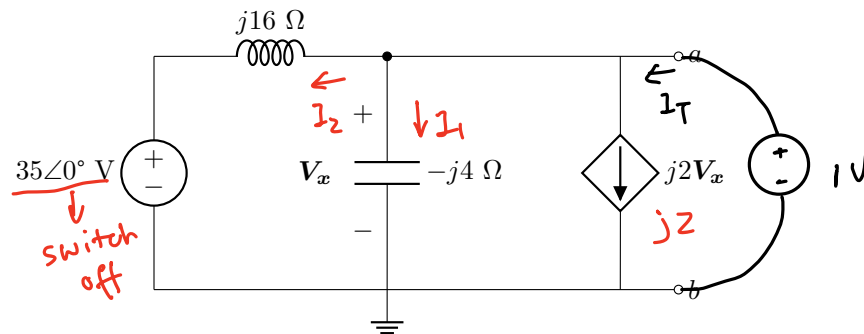


$$I = j2 - j1.6 = j0.4$$

$$Z = \frac{j5 \times 10}{10 + j5} = 2 + j4 \text{ } \Omega$$

Questions 23 to 24 [4]

Calculate the Thévenin equivalent voltage V_{Th} , and the Thévenin equivalent impedance Z_{Th} , with respect to terminals $a - b$.



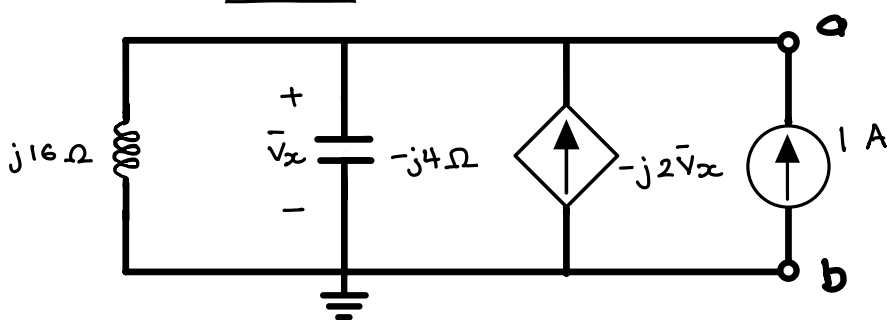
23	V_{Th} [2]	-1 V
24	Z_{Th} [2]	$-j 0.4571 \Omega$

For $\bar{V}_{Th}$

$$\begin{aligned} \frac{\bar{V}_x - 0}{-j4} + \frac{\bar{V}_x - 35}{j16} + j2\bar{V}_x &= 0 \\ j4\bar{V}_x - j(\bar{V}_x - 35) + j32\bar{V}_x &= 0 \\ j4\bar{V}_x - j\bar{V}_x + j32\bar{V}_x + j35 &= 0 \\ j35\bar{V}_x + j35 &= 0 \\ \underline{\underline{\bar{V}_x = -1 \text{ V}}} \end{aligned}$$

$$\begin{aligned} V_x &= 1, \\ I_T &= j2 + \frac{1}{j16} + \frac{1}{-j4} \\ &= j2.1875 \\ Z_T &= \frac{1}{I_T} = -j0.4571 \end{aligned}$$

For $\bar{Z}_{Th}$:

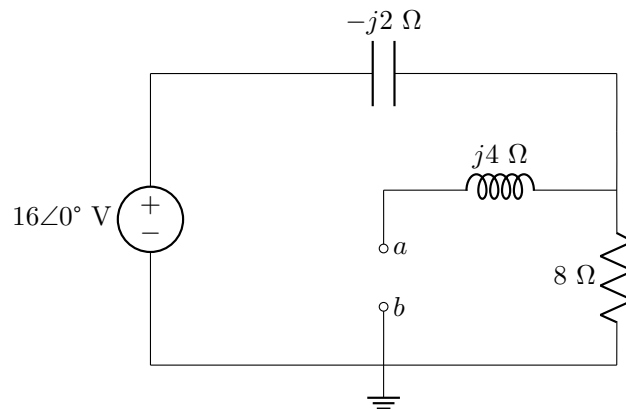
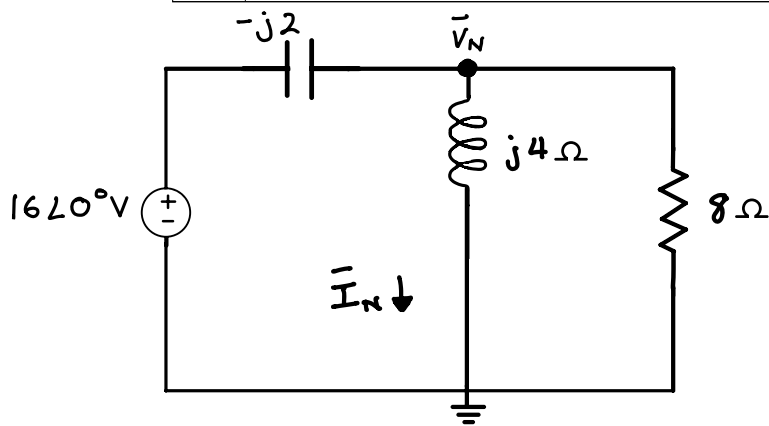


$$\bar{Z}_{Th} = \frac{\bar{V}_{ab}}{\bar{I}_{ab}} = \frac{\bar{V}_{ab}}{1}$$

$$\bar{V}_{ab} = \bar{V}_x = (-j0.4571) \text{ V}$$

$$\bar{Z}_{Th} = -j0.4571 \Omega$$

$$\begin{aligned} \frac{\bar{V}_x - 0}{j16} + \frac{\bar{V}_x - 0}{-j4} + j2\bar{V}_x - 1 &= 0 \\ -j\bar{V}_x + j4\bar{V}_x + j32\bar{V}_x - 16 &= 0 \\ j35\bar{V}_x - 16 &= 0 \\ j35\bar{V}_x &= 16 \\ \bar{V}_x &= \frac{16}{j35} \\ \bar{V}_x &= \underline{\underline{-j0.4571}} \end{aligned}$$

Question 25 [2]Determine the Norton equivalent current I_N with respect to terminals $a - b$.25 | I_N [2] | 3.2 - j6.4 A

$$\frac{\bar{V}_N - 16\angle 0^\circ}{-j2} + \frac{\bar{V}_N - 0}{j4} + \frac{\bar{V}_N - 0}{8} = 0$$

$$j4[\bar{V}_N - 16] - j2\bar{V}_N + \bar{V}_N = 0$$

$$j4\bar{V}_N - j64 - j2\bar{V}_N + \bar{V}_N = 0$$

$$(1 + j2)\bar{V}_N = j64$$

$$\bar{V}_N = \frac{j64}{1 + j2}$$

$$\bar{V}_N = 25.6 + j12.8 \text{ V}$$

$$\bar{I}_N = \frac{\bar{V}_N}{j4}$$

$$\bar{I}_N = \frac{25.6 + j12.8}{j4}$$

$$\bar{I}_N = 3.2 - j6.4 \text{ A}$$

Total Section 3: [18]**Grand Total : [46]**

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	3	E	B	N	1	1	1	S	0	3
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Please round all irrational numbers to a minimum of 5 significant digits.

Answer		Unit
Section 1		
01	$v_C(t = 1 \text{ s}) =$	
02	$v_L(t = 1 \text{ s}) =$	
03	$w_{C_{total}} =$	
04	$w_{L_{total}} =$	
05	$C_{ab} =$	
Section 2		
06	$Z = -0.5 + j0.25$	No unit required
07	$I_1 = -0.7089 + j2.9915$	No unit required
08	$\theta = 105$	Degree
09	$I = 16.213 - j29.873$	No unit required
10	$Z_{ab} = 54.4 + j40.8$	Ω
11	$I = j2.828$	A
12	$I_1 = 5.3846 - j3.0769$	A
13	$V_2 = -48 - j96$	V
14	$I_o = -1.28 + j2.56$	A
Section 3		
15	$I_x = j0.3261$	A
16	$a = -3$	No unit required
17	$b = -1$	No unit required
18	$c = 1 + j$	No unit required
19	$d = -6 - j162$	No unit required
20	$I_x^V = -10$	A
21	$I = j8.4$	A
22	$Z = 2 + j4$	Ω
23	$V_{Th} = -1$	V
24	$Z_{Th} = -j0.4571$	Ω
25	$I_N = 3.2 - j6.4$	A